

Threat Score Calculation Engine: Technical Specification & Mathematical Formulation

Document Version: 2.4.0	Classification: Proprietary SOC Architecture	Target Pipeline: AMVTF Multi-Signal Fusion	Storage Model: Strictly Local SQLite
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1. Executive Overview & Multi-Modal Architecture

VoxGuard calculates real-time threat scores ranging from **0.0 (Completely Safe)** to **100.0 (Critical Threat)**. Unlike legacy voice systems that rely on a single acoustic checkpoint, VoxGuard deploys a **Tri-Engine Triangulation Pipeline**. Audio signals, biometric identity profiles, and conversational transcripts are evaluated simultaneously by three dedicated neural engines before being passed to the **AMVTF (Adaptive Multi-Signal Voice Trust Fusion)** master engine for final verdict synthesis.

Sub-Engine	Component Name	Primary Analyzed Metric	Threat Conversion
Acoustic Engine	AASIST GNN Audio Physics	Glottal jitter, vocoder aliasing, pause cadence, speaker replay	Acoustic Threat = 100 - Authenticity
Identity Engine	ECAPA-TDNN Biometrics	192-dim acoustic embeddings cosine similarity vs enrolled profiles	Identity Threat = 100 - Identity
Contextual Engine	NLP Intent & Threat Classifier	10-category scammer phrasing, OTP theft, coercion, urgency	Context Threat = contextRisk
Master Fusion	AMVTF v2.4 Fusion Engine	Dynamic contextual weighting based on session mode and deepfake risk	Fused Threat Score (0 - 100)

2. Acoustic Authenticity Analysis (AASIST Engine)

The AASIST engine (backend/engines/aasist.py) conducts biological acoustic physics checks at 16kHz. Human speech naturally contains unvoiced sibilants ('s', 'sh', 'f', 't'). AASIST strictly extracts Voiced Frame Resonances (F0) to prevent false positives from background room noise or microphone handling taps.

Physical Artifact / Defect	Detection Mechanism	Anomaly Threshold	Threat Penalty
Robotic Pitch Rigidity	Real humans have neuromuscular micro-tremor. TTS is unnaturally flat.	Jitter < 0.18% OR (Pitch CV < 2% & Range < 5Hz)	+45.0 points
Isochronous Word Pacing	AI speech models generate uniform, equidistant inter-word pauses.	Pause Standard Deviation < 10ms across 3+ pauses	+35.0 points
Neural Vocoder Aliasing	Transposed 1D convolutions cause mirror energy leakage > 6.5kHz.	Voiced High-Band / Mid-Band Energy Ratio > 0.360	+30.0 to +45.0 pts
Phase Discontinuity	Vocoders (e.g., ElevenLabs) exhibit phase glitches with smooth kurtosis.	Jitter > 3.0% AND Hilbert Envelope Kurtosis < 13.5	+35.0 points
Loudspeaker Replay Cutoff	AI voice played over phone speaker shows hardware bass roll-off.	Sub-220Hz Energy / Mid-Band Ratio < 0.025	+30.0 points

Bona Fide Human Voice Verification: If jitter is within [1.20%, 2.85%], envelope kurtosis ≥ 13.0, pitch range ≥ 15Hz, and voiced HF ratio < 0.35, all biometric signatures confirm a living human speaker. Threat penalties are cleared to 0.0.

- **Bona Fide Human Authenticity:** authenticity = clip(94.0 - penalty * 0.10, 88.0, 97.5)
- **Synthetic Voice Detected:** authenticity = clip(42.0 - penalty * 0.30, 12.0, 38.0)

3. Speaker Identity Biometrics (ECAPA-TDNN Engine)

The ECAPA-TDNN engine (backend/engines/ecapa.py) projects audio frames across pseudo-mel filterbanks to synthesize a normalized **192-dimensional acoustic embedding vector**. Cosine similarity is computed against enrolled SQLite profiles:

Cosine Similarity Formulation: $\text{Similarity \%} = \text{clip}((\text{Live_Embedding} \cdot \text{Enrolled_Embedding}) / (||\text{Live}|| \times ||\text{Enrolled}||) \times 100, 0.0, 100.0)$

- **Verification Threshold:** $\geq 70.0\%$ similarity constitutes a confirmed voiceprint match.
- **Identity Threat Penalty:** When target profile verification is actively requested, $\text{Identity Threat} = 100.0 - \text{SimilarityScore}$.

4. Contextual Threat & Intent NLP Engine

The Contextual Threat Engine (backend/engines/nlp.py) inspects the spoken transcript tokens for coercive fraud, social engineering, credential theft, and psychological pressure across 10 specialized categories:

Threat Category	Base Weight	Addl Match	Example Keywords & Social Engineering Tactics
OTP / Verification Fraud	45.0	+15.0	tell me the otp, share verification code, 6 digit code, sms code, read the otp
Credential & PII Theft	42.0	+14.0	cvv number, card expiry, atm pin, install anydesk, teamviewer, screen share
Card / Account Expiry	40.0	+12.0	card will expire, account suspended, kyc expired, update kyc, link pan card
Fear & Legal Extortion	40.0	+12.0	arrest warrant, police complaint, court notice, don't hang up, stay on the line
Financial Manipulation	38.0	+12.0	upi pin, scan this qr, send money, refund process, lottery prize, double money
Security Protocol Bypass	35.0	+12.0	bypass protocol, skip dual-auth, disable mfa, override security, skip verification
Impersonation Tactics	35.0	+10.0	calling from bank, reserve bank rbi, cyber crime, customs, tech support, sbi, hdfc
Sensitive Asset Targeting	30.0	+10.0	wire transfer, routing number, swift code, lost 2fa device, send recovery link
Artificial Urgency Pressure	20.0	+8.0	immediately, within 5 minutes, act now, hurry, asap, emergency, expires today
Executive Authority	18.0	+6.0	ceo requested, board meeting, cfo authorized, executive order, confidential project

- **Multi-Vector Attack Escalation:** If phrases trigger across ≥ 3 categories simultaneously, a **+15.0 bonus** is added.
- **Context Risk Final Score:** $\text{contextRisk} = \text{clip}(\text{raw_score} + 10.0, 0.0, 100.0)$. If no triggers: baseline is 8.0.

5. AMVTF Master Fusion Engine & Threat Formulas

The AMVTF engine (backend/engines/amvtf.py) synthesizes all three threat signals into the unified **Fused Threat Score**. The fusion weighting adapts dynamically to the security context:

Scenario A: Target Profile Verification Mode (Specific Identity Claimed)

When an incoming call claims to represent a specific enrolled executive or customer, all 3 signals participate in tri-fusion:

Formula: $\text{Fused Threat Score} = (0.40 \times \text{Acoustic_Threat}) + (0.35 \times \text{Identity_Threat}) + (0.25 \times \text{Context_Threat})$
• 40% Acoustic Authenticity • 35% Voiceprint Biometric Match • 25% Contextual Intent

Scenario B: Live General Monitoring Mode (Unenrolled / Unknown Caller Intake)

When monitoring live calls where no specific target identity is claimed, fusion adapts according to the caller's acoustic authenticity:

- **Case 1: Living Human Voice (Authenticity $\geq 80.0\%$):**
Acoustic baseline is clean (~5-10% risk). Threat is predominantly driven by spoken fraud keywords:
 $\text{Fused Score} = (0.15 \times \text{Acoustic_Threat}) + (0.85 \times \text{Context_Threat})$
- **Case 2: Synthetic Voice / Neural Vocoder Detected (Authenticity $< 80.0\%$):**
Acoustic anomalies present immediate threat of automated AI voice clone attack:
 $\text{Fused Score} = (0.65 \times \text{Acoustic_Threat}) + (0.35 \times \text{Context_Threat})$

6. Security Verdicts, Action Types & Automated SOC Responses

Verdict	Decision Thresholds	Action Type	SOC Security Response
CRITICAL	- Authenticity < 50.0% (AI Deepfake) - Context Risk ≥ 70.0% (Active Fraud) - Fused Threat Score ≥ 60.0	TERMINATE	TERMINATE CALL IMMEDIATELY / BLOCK TRANSACTION Automated session cutoff. Active coercion / voice synthesis alert dispatched to SOC SIEM.
WARNING	- Fused Threat Score ≥ 35.0 - Context Risk ≥ 38.0% - Voiceprint Mismatch (< 70%)	STEP_UP	Trigger Out-of-Band (OOB) Secondary Verification Hold high-risk action. Require secondary multi-factor challenge (SMS OTP / hardware token).
SAFE	- Fused Threat Score < 35.0 - Authenticity ≥ 50.0% - Context Risk < 38.0%	ALLOW	Allow Session — Normal Operations Bona fide human voice verified. Natural pitch modulation confirmed. No fraud keywords detected.

7. Step-by-Step Numerical Walkthrough Examples

Case 1: Routine Operational Business Call (Benign Human Voice)

Input: Employee calling desk to query standard schedule. No profile requested.

- Acoustic:** Jitter = 1.85%, Kurtosis = 16.2, Voiced HF = 0.18 \$implies\$ **Authenticity = 94.0%** (Acoustic Threat = 100 - 94 = 6.0)
- Transcript:** *'Hi, could you send over the updated project roadmap when ready?'* \$implies\$ **Context Risk = 8.0**
- Fusion:** Human voice rule applied: Fused Score = (0.15 × 6.0) + (0.85 × 8.0) = 0.9 + 6.8 = 7.7
- SOC Decision:** **SAFE** (Action: ALLOW). Session allowed with zero operational friction.

Case 2: Social Engineering OTP Scam (Living Human Voice)

Input: Fraudster calling elderly banking customer. Living human acoustic baseline.

- Acoustic:** Biological voice verified \$implies\$ **Authenticity = 92.0%** (Acoustic Threat = 100 - 92 = 8.0)
- Transcript:** *'This is bank security. Your debit card is expired and account is blocked. Share the 6 digit OTP immediately!'*
 - Matches: Impersonation (35) + Card Expiry (40) + OTP Fraud (45) + Urgency (20) + Multi-Vector Bonus (+15) + Offset (+10) \$implies\$ **Context Risk = 100.0**
- Fusion:** Human voice rule applied: Fused Score = (0.15 × 8.0) + (0.85 × 100.0) = 1.2 + 85.0 = 86.2
- SOC Decision:** **CRITICAL** (Action: TERMINATE). Call intercepted and blocked immediately.

Case 3: Automated AI Voice Clone / Deepfake Impersonation

Input: ElevenLabs synthesized voice targeting wire-transfer desk.

- Acoustic:** Neural vocoder aliasing (voiced HF = 0.420) + flat pitch (range = 3.8Hz) \$implies\$ **Authenticity = 22.0%** (Acoustic Threat = 100 - 22 = 78.0)
- Transcript:** *'Good morning, following up on the emergency payment transfer requested earlier.'* \$implies\$ **Context Risk = 38.0**
- Fusion:** Deepfake suspect rule applied: Fused Score = (0.65 × 78.0) + (0.35 × 38.0) = 50.7 + 13.3 = 64.0
- SOC Decision:** **CRITICAL** (Action: TERMINATE). Synthetic deepfake voice detected.